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Countercurrent staged diafiltration for formulation of high value proteins

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Abstract

A number of groups have studied the application of continuous bioreactors and continuous chromatographic systems as part of efforts to develop an integrated continuous biomanufacturing process. The objective of this study was to examine the feasibility of using a countercurrent staged diafiltration process for continuous protein formulation with reduced buffer requirements. Experiments were performed using a polyclonal immunoglobulin (IgG) with Cadence™ Inline Concentrators. Model equations were developed for the product yield, impurity removal, and buffer requirements as a function of the number of stages and the stage conversion (ratio of permeate to feed flow rate). Data from a countercurrent two-stage system were in excellent agreement with model calculations, demonstrating the potential of using countercurrent staged diafiltration for protein formulation. Model simulations demonstrated the importance of the countercurrent staging on both the extent of buffer exchange and the amount of buffer required per kg of formulated product. The staged diafiltration process not only provides for continuous buffer exchange, it could also provide significant reductions in the number of pump passes while providing opportunities for reduced buffer requirements.

KEYWORDS
antibody, continuous processing, diafiltration, formulation, ultrafiltration

1 | INTRODUCTION

There is growing interest in the development of continuous processes for the production of high value biopharmaceuticals due to the potential for greater productivity, lower capital costs, enhanced product quality, and greater manufacturing flexibility (Zydney, 2016). The use of perfusion bioreactors for continuous production of labile enzymes and monoclonal antibody products is already well established (Konstantinov et al., 2006). A number of recent studies have explored the development of continuous chromatographic processes using both multicolumn systems (Bryntesson, Hall, & Lacki, 2011; Godawat et al., 2012) and countercurrent tangential chromatography (Dutta et al., 2015). Single pass tangential flow filtration systems with high conversion (ratio of permeate to feed flow rate greater than 50%) can provide continuous concentration of monoclonal antibody products (Casey, Gallos, Alekseev, Ayturk, & Pearl, 2011).

However, there has been very little work on the development of continuous diafiltration processes. Diafiltration is used for the final formulation of essentially all biopharmaceuticals products and it can also provide preconditioning of the feed stream prior to other purification steps in the downstream process (Van Reis & Zydney, 2007). Current diafiltration processes are inherently batch operations, with the feed recirculated through the membrane module while diafiltration buffer is continuously added and permeate is removed (Cheryan, 1998).

Several recent studies have explored the feasibility of continuous diafiltration processes. For example, Mohanty and Ghosh (2008) showed that a tangential flow ultrafiltration membrane cascade with a countercurrent configuration could be used to continuously purify a

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monoclonal antibody (Campath-1H) from a simulated mammalian cell culture supernatant, although the maximum purification factor obtained in this work was less than 12 compared to the 1000-fold impurity removal typically targeted in most diafiltration systems for bioprocessing applications. Lin and Livingston (2007) demonstrated the use of a countercurrent nanofiltration membrane cascade for organic solvent exchange, although the extent of buffer exchange was only about 4-fold (corresponding to 77% exchange). Lipnizki, Boelsmand, & Madsen, (2002) presented limited results for a continuous countercurrent process for protein diafiltration based on earlier work by Madsen (2001) that showed the potential for reduced buffer consumption due to the countercurrent staging, although no quantitative analysis of the system performance was presented. Lutz (2015) described the use of a single-pass diafiltration process with diafiltration buffer added between stages (equivalent to repeated steps of concentration and dilution). Reference was also provided to work by Westoby on the development of a countercurrent staged diafiltration process that is analogous to the system analyzed in this study (Lutz, 2015).

The objective of this work was to demonstrate the feasibility of using countercurrent staged diafiltration for buffer exchange/formulation of protein products. Data were obtained using polyclonal (serum) IgG as a model product with vitamin B_{12} as a low molecular weight impurity, with the diafiltration performed using Cadence™ Inline Concentrators that can provide high conversion in single pass operation. Results were in excellent agreement with model calculations, showing the importance of the countercurrent staging on both the extent of buffer exchange and the buffer requirements for the diafiltration process.

2 | MATERIALS AND METHODS

Countercurrent staged diafiltration was performed with Cadence™ Inline Concentrators provided by Pall Corporation (Port Washington, NY) with Delta regenerated cellulose membranes having a nominal molecular weight cutoff of 30 kDa and 650 cm^2 of membrane area. Each module was flushed with 2 L of deionized water at a feed pressure of approximately 30 psi (210 kPa) followed by 10 mM acetate buffer at pH 5; acetate is one of the most commonly used buffers for formulation of therapeutic antibodies (Kang, Lin, & Penera, 2016). Feed and retentate flow rates were controlled using Masterflex L/S peristaltic pumps (Cole-Parmer, Vernon Hills, IL) fitted with platinum-cured silicone tubing (Cole-Parmer). Pumps were calibrated before each experiment by timed volume collection using a digital balance (OHAUS, Pine Brook, NJ). Pressures were monitored using stainless steel pressure gauges (Ashcroft, Stratford, CT).

Serum immunoglobulin G (IgG) from SeraCare (Milford, MA) was used as the model protein feed, with vitamin B_{12} (cyanocobalamin, 1355 Da, Sigma-Aldrich, St. Louis, MO), used as a model impurity in all experiments. IgG and vitamin B_{12} were dissolved in 10 mM acetate buffer at pH 5. All solutions were pre-filtered through 0.2 µm Supor® 200 filters (Pall, Ann Arbor, MI) prior to use. The solution conductivity was measured using a 105APlus Conductivity meter (Thermo Orion, Beverly, MA), and the pH was measured using a 420APlus pH meter (Thermo Orion).

Vitamin B_{12} and IgG concentrations were determined using a NanoDrop 2000c Spectrophotometer (Thermo Fisher Scientific, Waltham, MA) based on the absorbance at 550 and 280 nm, respectively. Calibration curves were constructed using mixtures containing known concentrations of B_{12} and IgG. The dilute concentrations of vitamin B_{12} provided negligible contribution to the absorbance at 280 nm (compared to IgG). The B_{12} concentrations were then determined from the absorbance at 550 nm by subtracting off the contribution from IgG at this wavelength.

All experiments were performed with constant feed/permeate flow rates. 500 µl samples were obtained from the retentate exit at specified time intervals for off-line evaluation of the IgG and B_{12} concentrations. At the end of the experiment, the entire system was flushed with buffer and then cleaned with 2 L of 0.25 N NaOH. When required, the membranes were back-flushed using a small volume of NaOH. The modules were then capped and stored in 0.25 N NaOH between experiments.

3 | RESULTS AND ANALYSIS

3.1 | Two-stage diafiltration

Figure 1 shows a schematic of the two-stage countercurrent diafiltration system. The protein feed is combined with the permeate from Stage 2, with the “diluted” protein solution fed to Stage 1. The retentate from Stage 1 is then mixed with fresh diafiltration buffer and fed back to Stage 2, with the diafiltered product solution recovered in the retentate. The countercurrent staging effectively dilutes and concentrates the feed twice, enabling a much higher level of impurity removal (or buffer exchange) than would be possible with just a single stage system. The flow rate of the diafiltered product was controlled by a metering valve (Swagelok , Pittsburgh, PA), while the other flow rates were controlled using Masterflex pumps.

Figure 2 shows the IgG and vitamin B_{12} concentrations in the exit stream from Stage 2 (the diafiltered product) as a function of time using the two-stage countercurrent diafiltration system shown in Figure 1. The system was operated with a feed flow rate of q_F = 2 ml/min and a diafiltration buffer flow rate of q_{DF} = 18 ml/min, with both Inline Concentrators providing 90% conversion, corresponding to a filtrate flux of 4.6 µm/s = 17 L/m^2/hr. The transmembrane pressure in each module was approximately 24 kPa = 3.5 psi, with the pressure drop across the module (feed inlet minus retentate exit) being approximately 48 kPa = 7 psi. The initial feed volume was 280 ml with an IgG concentration of 106 ± 6 g/L.

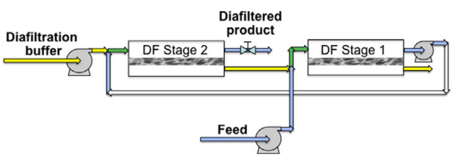

FIGURE 1 Schematic of two-stage countercurrent diafiltration system

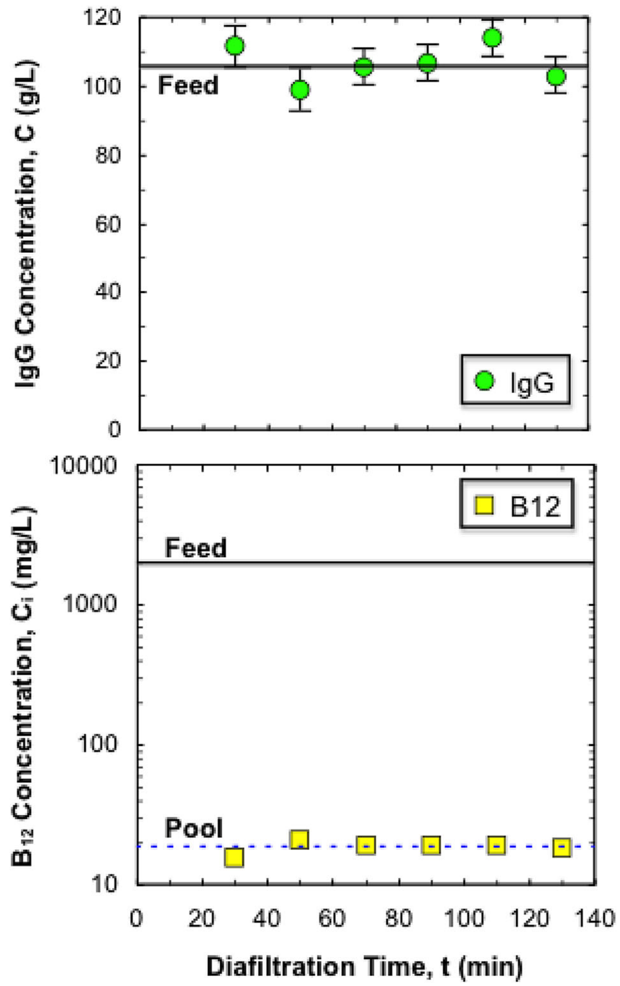

FIGURE 2 IgG (top panel) and vitamin B_{12} (bottom panel) concentrations as a function of time during a two-stage countercurrent diafiltration performed with q_F = 2 ml/min and q_{DF} = 18 ml/min. Error bars on IgG concentrations determined from accuracy of the concentration measurements. Error bars for vitamin B_{12} values are within the size of the symbols and are not shown

The IgG concentration in the diafiltered product (top panel) was essentially constant throughout the diafiltration process (for t > 20 min), with the final concentration equal to 111 ± 6 g/L, corresponding to 104 ± 5% yield. There was an initial time lag at the start of the process associated with the washout of the hold-up volume in the modules (approximately 7 ml for each module) and the tubing/pressure gauges (approximately 28 ml total). The first 30 ml of output was simply discarded. The B_{12} concentration was also stable after this initial time lag; the B_{12} concentration at the start of the process was well below 20 mg/L since the module was initially filled with only acetate buffer. The B_{12} concentration in the diafiltered product was 19 ± 1 mg/L compared to the feed concentration of 1810 ± 50 mg/L, corresponding to R = 95 ± 8 (equivalent to an impurity removal of 98.9 ± 0.1%). The performance of the diafiltration process was stable throughout the 130 min process, corresponding to a total challenge of 0.21 kg of protein per m^2 of total membrane area for the two Inline Concentrators. The total amount of buffer used for the diafiltration was 85 L/kg of IgG. Note that a typical diafiltration process conducted using a feed concentration of 60 g/L, which is equal to the mean concentration in the modules used in these experiments (10 g/L inlet, 110 g/L outlet), would use 130 L/kg for an eight diavolume process, while a process giving R = 95 would use 75 L/kg (Van Reis & Zydney, 2007). The buffer requirements in both the continuous and batch diafiltrations could be reduced by operating at a higher protein concentration; this is discussed in more detail subsequently.

3.2 | Model calculations

The degree of impurity removal, or buffer exchange, for a countercurrent staged diafiltration process can be evaluated by solving the governing mass balance equations for each stage, analogous to the approach used for classical staged unit operations like mixer-settler combinations for extraction (Harrison, Todd, Rudge, & Petrides, 2002), giving:

$$R = \frac{\alpha^{N+1} - 1}{\alpha - 1} \tag{1}$$

where N is the number of stages and $R = C_F/C_N$, with the subscripts F and N referring to the feed and the final diafiltered product (output from Nth stage). The flow removal factor α is given as:

$$\alpha = \frac{q_{DF} S}{q_F} \tag{2}$$

where q_{DF} is the volumetric flow rate of the diafiltration buffer, q_F is the feed flow rate, and S is the solute sieving coefficient, equal to the solute concentration in the permeate stream divided by that in the feed. A small, non-retained, solute will have S = 1. Note that Equation (1) assumes perfect mixing at the inlet to each stage with no loss of product (or impurity) due to adsorption to the membrane. Thus, Equation (1) predicts R = 1 (100% recovery) for the fully retained IgG product since S = 0. Equation (1) also neglects the effects of impurity binding to the product; this has been discussed in the context of batch diafiltration by Shao and Zydney (2004). Note that if the binding isotherm is linear, which is often a good approximation for dilute impurities, the effects of impurity binding can be modeled using an effective sieving coefficient that is equal to the fraction of unbound (free) impurity (Shao & Zydney, 2004). Thus, product-impurity binding would reduce the magnitude of the flow removal factor α, thereby reducing the extent of impurity removal.

The experimental system examined in Figure 2 was operated with α = 9 (assuming no retention of the small vitamin B_{12} by the 30 kDa membrane), giving R = 91 for the two-stage countercurrent diafiltration. This is in excellent agreement with the value of R = 95 ± 8 determined experimentally. This provides strong evidence that the assumptions of perfect mixing, negligible adsorption, and no impurity binding are appropriate for this system. Based on Equation (1), the degree of impurity removal could be increased to R = 820 by using a 3-stage system and to 7,400 for a 4-stage system using a constant flow removal factor of α = 9. Alternatively, the 2-stage system could provide R = 1,000 by increasing

TABLE 1 Impurity removal and buffer requirements for 2-stage countercurrent diafiltration. Error bars on experimental values determined from accuracy of concentration measurements using standard propagation of error analysis

C_F (g/L)	q_F (ml/min)	Flow removal, α	*R* (expt)	*R* (theory)	Buffer, V_{DF} (L/kg)
130	2.0	4	26 ± 4	21	30
45	5.0	9	91 ± 6	91	200
110	2.0	9	95 ± 8	91	90
90	2.5	19	350 ± 40	381	210

the flow removal factor to α = 31, although this high a conversion might well be difficult to achieve using the existing Inline Concentrators.

The amount of buffer required for the diafiltration process is given as:

$$V_{DF} = \frac{\alpha}{C_F S} \tag{3}$$

where V_{DF} is the buffer volume required per kg of product. In sharp contrast to a batch diafiltration, in which the required buffer volume increases linearly with the number of diavolumes (which controls the degree of buffer exchange/impurity removal), V_{DF} in the countercurrent staged diafiltration is independent of the number of stages since the "same" buffer is simply cycled through the multiple stages in the countercurrent staged system. However, for a fixed number of stages, an increase in the degree of buffer exchange would require an increase in the flow removal factor (α) which could be achieved by increasing the dilution of the initial feed. The relative advantages/disadvantages of the staged diafiltration process are discussed in more detail subsequently.

3.3 | Diafiltration performance

Experimental results from several 2-stage countercurrent diafiltration runs are summarized in Table 1. These examined a range of feed concentrations and flow removal factors (conversions), with the feed flow rate ranging from q_F = 2.0–5.0 ml/min (with q_{DF} as high as 47.5 ml/min for the run with α = 19). The error bars represent the calculated errors determined from the accuracy of the vitamin B_{12} concentration measurements using standard propagation of error analysis. The degree of impurity removal evaluated experimentally was in excellent agreement with the theoretical predictions given by Equation (1) for all four experiments. The impurity removal increases with increasing flow removal factor (increasing conversion) due to the greater dilution factor in each stage. The experiment using a feed IgG concentration of 130 g/L with α = 4 required only 30 L/kg of buffer, although this system provided relatively little impurity removal. However, it would have been possible to obtain *R* > 1,000 using a 6-stage system with otherwise identical conditions.

The performance of the countercurrent staged diafiltration process is examined in more detail in Figure 3. The degree of impurity removal (or buffer exchange) is plotted as a function of the flow removal factor for systems with 1, 2, 3, and 4 stages. The symbols are the experimental data from Table 1. The results with a single stage system are equivalent to simple dilution (and re-concentration) of the feed; thus, *R* = 10 for α = 9 and *R* = 40 for α = 39. High degrees of impurity removal are impossible (from a practical perspective) with a single stage system due to the very high conversion (dilution) required, but the results with *N* = 4 show more than a 10,000-fold reduction in the impurity concentration (or degree of buffer exchange) with α > 9.8, which is easily achieved using the Inline Concentrators examined in this study. Note that low levels of buffer exchange can be accomplished using only a 2-stage system with moderate values of α. This might well be sufficient for applications of diafiltration to "pre-condition" a feed prior to a chromatographic process, for example, for reduction of ionic strength prior to ion exchange chromatography.

Figure 4 shows the buffer requirements (in L of buffer per kg of product) as a function of the impurity removal for the countercurrent staged diafiltration process for systems with 2, 3, 4, and 5 stages. In each case, the value of the flow removal factor (α) was calculated for the degree of impurity removal using Equation (1), with the buffer requirement then evaluated using Equation (3) assuming no impurity retention (*S* = 1) and a feed concentration of 100 g/L. The symbols are the experimental results for the two-stage system. The small discrepancies between the data and model are due in large part to the actual feed concentration in the diafiltration experiments being

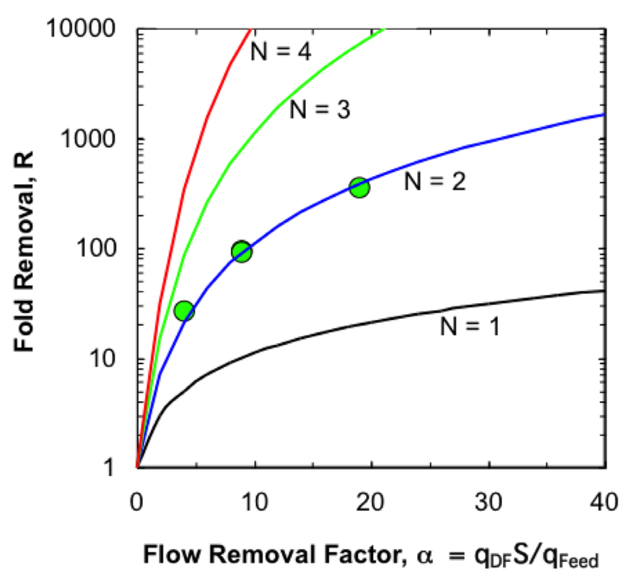

FIGURE 3 Impurity removal as a function of the flow removal factor for countercurrent staged diafiltration processes with *N* = 1, 2, 3, and 4. Error bars are within the size of the symbols and are not shown

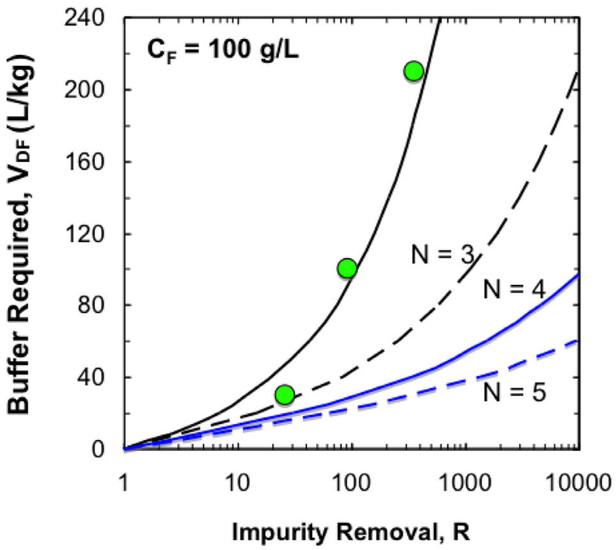

FIGURE 4 Buffer requirements (in L of buffer per kg of product) as a function of impurity removal for countercurrent staged diafiltration processes with two to five stages. Error bars are within the size of the symbols and are not shown

somewhat different than the 100 g/L used in the model. The volume of buffer needed for the diafiltration increases significantly with increasing degree of buffer exchange, particularly at large values of *R*. Increasing the number of stages provides a very large reduction in buffer requirements, with V_{DF} decreasing from 1000 L/kg for a 2-stage system with 10,000-fold impurity removal to only 61 L/kg for a 5-stage system with the same degree of buffer exchange.

The minimum buffer requirement for the countercurrent staged diafiltration process can be evaluated from Equations (1) and (3) by taking the limit as N → ∞, giving:

$$V_{min} = \frac{1 - \frac{1}{R}}{C_F S} \tag{4}$$

Equation (4) reduces to $V_{min} = 1/C_F$ for a diafiltration process that provides a very high degree of impurity removal ($R \to \infty$) with no impurity retention ($S = 1$). This gives $V_{min} = 10$ L/kg for a countercurrent diafiltration process performed using a feed concentration of 100 g/L. The corresponding amount of buffer required for a batch diafiltration process is:

$$V_{DF} = \frac{N_D}{C_F} \tag{5}$$

where N_D is the number of diavolumes. Equation (5) can be rewritten in terms of the degree of impurity removal as (Van Reis & Zydney, 2007):

$$V_{DF} = \frac{ln(R)}{C_F S} \tag{6}$$

In sharp contrast to the countercurrent staged diafiltration, the batch process would require an infinite volume of buffer to achieve an infinite degree of impurity removal ($R \to \infty$). A batch diafiltration with 10,000-fold impurity removal requires 150 L/kg if the diafiltration is performed with a product concentration of 60 g/L and 90 L/kg if using a product concentration of 100 g/L (as in the previous calculations for the staged system); these are both well above the 10 L/kg minimum requirement for the countercurrent staged diafiltration. Note that further increases in *R* would require a corresponding increase in buffer requirement for the batch diafiltration, which is in sharp contrast to the behavior of the countercurrent staged process where V_{min} becomes independent of *R* at large degrees of buffer exchange (Equation 4).

4 | CONCLUSIONS

The results presented in this study provide the first demonstration of the feasibility of using a countercurrent staged diafiltration process for formulation of high-value biological products like monoclonal antibodies. A two-stage system was able to provide nearly 400-fold removal of a small impurity by operating with a flow removal factor of α = 19, which is readily obtained using commercial Cadence™ Inline Concentrators. The experimental results from this two-stage system were in excellent agreement with model calculations accounting for the countercurrent staging, providing a simple framework for evaluating the number of stages and flow removal factor required for a given degree of buffer exchange.

The permeate flux for the experiment presented in Figure 2 was only 17 L/m^2/hr. However, we were able to obtain permeate flux values as high as 46 L/m^2/hr at a conversion of 90% using a feed concentration of 10 g/L (corresponding to an exit concentration of 100 g/L), which is typical of the filtrate flux in current diafiltration processes (Van Reis & Zydney, 2007). Operation with higher exit concentrations (e.g., 200 g/L) led to fouling, most likely associated with the high protein concentrations at the membrane surface (which could exceed the gel concentration) along with the high viscosity of the IgG solution under these conditions (Binabaji, Ma, & Zydney, 2015). Note that if the staged and batch diafiltration processes are operated with the same permeate flux, the continuous staged process would take less time than the batch process since it requires less buffer (and thus less permeate volume) to achieve the same degree of buffer exchange. It is not currently possible to predict the permeate flux in the Cadence™ Inline Concentrators based on the device operating conditions due to the complexity of the modules (internal stages with changing number of parallel channels) along with the change in retentate flow rate and protein concentration associated with the high conversion in these modules.

There are a number of potential advantages to the proposed countercurrent staged diafiltration system compared to the traditional batch diafiltration process currently used for buffer exchange, both for final product formulation and preconditioning of feed streams. First, the staged diafiltration can be operated continuously, providing opportunities for direct integration into a continuous biomanufacturing process. Second, the staged diafiltration process operates in a single pass mode, so that the product experiences only *N* pump passes (where *N* is the number of stages). This is in sharp contrast to a batch diafiltration process where the product can be circulated through the feed pump 10–100 times over the course of the diafiltration. Third, the buffer requirements for the staged diafiltration process can be reduced to below that for an equivalent batch process by increasing the number of stages since the buffer requirement is essentially independent of N. However, it is important to note that the staged diafiltration process requires a considerably more complex system, it employs a greater number of pumps (increasing the capital cost), and it uses a greater amount of membrane area. There may also be challenges maintaining stable flow rates throughout the process. Future studies will be required to identify the most appropriate opportunities, and the optimal design configurations, for implementation of this type of countercurrent staged diafiltration in biomanufacturing.

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